Laws of sets: Commutative, associative and distributives

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1 Introduction

In set theory, several fundamental laws govern how sets interact with each other. These laws help in understanding and manipulating sets. Here's a detailed overview of each law, with examples.

2 Commutative laws

The commutative laws state that the order in which two sets are combined does not affect the result.

2.1 Union:

Let A and B be two sets, then:

$$A \cup B = B \cup A$$

Example:

If $A = \{1, 2\}$ and $B = \{2, 3\}$, then

$A \cup B = \{1, 2, 3\}$ and $B \cup A = \{1, 2, 3\}$.

Both sets are equal.

2.2 Intersection:

Let A and B be two sets, then

$$A \cap B = B \cap A$$

Example:

If $A = \{1, 2\}$ and $B = \{2, 3\}$, then

$A \cap B = \{2\}$ and $B \cap A = \{2\}$.

Both sets are equal.

3 Associative laws

The associative laws state that the grouping of sets does not affect the result when performing union or intersection.

3.1 Union:

Let A, B and C be three sets, then

$$(A \cup B) \cup C = A \cup (B \cup C)$$

Example:

If $A = \{1\}$, $B = \{2\}$, and $C = \{3\}$, then

$(A \cup B) \cup C = \{1, 2, 3\}$ and $A \cup (B \cup C) = \{1, 2, 3\}$.

Both sets are equal.

3.2 Intersection:

Let A, B and C be three sets, then

$$A \cap B) \cap C = A \cap (B \cap C)$$

Example:

If $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, and $C = \{3, 4, 5\}$, then

$(A \cap B) \cap C = \{3\}$ and $A \cap (B \cap C) = \{3\}$.

Both sets are equal.

4 Distributive laws

The distributive laws describe how union and intersection distribute over each other.

4.1 Union over intersection:

Let A, B and C be three sets, then

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Example:

If $A = \{1, 2\}$, $B = \{2, 3\}$, and $C = \{2, 4\}$, then

$A \cup (B \cap C) = \{1, 2\}$ and $(A \cup B) \cap (A \cup C) = \{1, 2\}$.

Both sets are equal.

4.2 Intersection over union:

Let A, B and C be three sets, then

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Example:

If $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, and $C = \{3, 4, 5\}$, then

$A \cap (B \cup C) = \{2, 3\}$ and $(A \cap B) \cup (A \cap C) = \{2, 3\}$.

Both sets are equal.

5 Identity laws

The identity laws state that combining a set with the universal set or the empty set yields the set itself.

5.1 Union with empty set:

Let A be a set, then

$$A \cup \emptyset = A$$

Example: If $A = \{1, 2\}$, then $A \cup \emptyset = \{1, 2\} = A$.

5.2 Intersection with universal set:

Let A be a set, then

$$A \cap U = A$$

Example: If $A = \{1, 2\}$ and U is the universal set, then $A \cap U = \{1, 2\} = A$.

6 Complement laws

The complement laws describe the relationship between a set and its complement. Please note that the notation A^c and $\bar{A}$ **both represent the same concept** in set theory and logic:

6.1 Union with complement:

Let A be a set and U the universal set, then

$$A \cup A^c = U$$

Example:

If $U = \{1, 2, 3\}$ and $A = \{1, 2\}$, then

$A^c = \{3\}$ and $A \cup A^c = \{1, 2, 3\} = U$.

6.2 Intersection with complement:

Let A be a set and U the universal set, then

$$A \cap A^c = \emptyset$$

Example:

If $U = \{1, 2, 3\}$ and $A = \{1, 2\}$, then

$A^c = \{3\}$ and $A \cap A^c = \emptyset$.

7 Absorption laws

The absorption laws simplify expressions by showing how combining a set with another set and their union or intersection affects the result.

7.1 Union:

Let A and B be two sets, then

$$A \cup (A \cap B) = A$$

Example:

If $A = \{1, 2\}$ and $B = \{2, 3\}$, then

$A \cap B = \{2\}$ and $A \cup (A \cap B) = \{1, 2\} = A$.

7.2 Intersection:

Let A and B be two sets, then

$$A \cap (A \cup B) = A$$

Example:

If $A = \{1, 2\}$ and $B = \{2, 3\}$, then

$A \cup B = \{1, 2, 3\}$ and $A \cap (A \cup B) = \{1, 2\} = A$.

8 Double negation laws

The double negation law states that taking the complement of a complement returns the original set.

Let A be a set and U the universal set, then

$$(A^c)^c = A$$

Example:

If $U = \{1, 2, 3\}$ and $A = \{1, 2\}$, then

$A^c = \{3\}$ and $(A^c)^c = \{1, 2\} = A$.

9 De Morgan's laws

De Morgan's laws relate the complement of the union of sets to the intersection of their complements, and vice versa.

9.1 Complement of union:

The complement of the union of two sets is equal to the intersection of their complements.

Let A, B and C be three sets, then

$$(A \cup B)^c = A^c \cap B^c$$

Example:

If $U = \{1, 2, 3, 4\}$, $A = \{1, 2\}$, and $B = \{2, 3\}$, then

$A \cup B = \{1, 2, 3\}$, $(A \cup B)^c = \{4\}$, and $A^c \cap B^c = \{4\}$.

Both sets contains the same elements.

9.1 Complement of intersection:

Let A, B and C be three sets, then

$$(A \cap B)^c = A^c \cup B^c$$

Example:

If $U = \{1, 2, 3, 4\}$, $A = \{1, 2\}$, and $B = \{2, 3\}$, then

$A \cap B = \{2\}$, $(A \cap B)^c = \{1, 3, 4\}$, and $A^c \cup B^c = \{1, 3, 4\}$.

10 Set difference laws

The difference of set A and set B, denoted as $A - B$ or $A \setminus B$, is the set of elements that are in A but not in B.

This can also be expressed using the intersection with the complement of B:

Let A and B be two sets, then

$$A - B = A \cap B^c$$

10.1 Example:

Let $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$. Then:

$$B^c = \{1\}$$
$$A \cap B^c = \{1, 2, 3\} \cap \{1\} = \{1\} \text{ and } A - B = \{1\}$$

As you can see both sets contain the same elements.

Understanding these laws of sets provides a strong foundation for working with sets in various areas of mathematics, computer science and logic. They help simplify complex set expressions and solve problems efficiently.

Exercise 1. Let's simplify the given expression using the laws of sets to simplify the expression $(A \cup B) \cap (A \cup B^c)$

Solution 1.

$$(A \cup B) \cap (A \cup B^c)$$

Step-by-step simplification

1. Distribute the intersection over the union:

$$(A \cup B) \cap (A \cup B^c) = (A \cap (A \cup B^c)) \cup (B \cap (A \cup B^c))$$

2. Simplify each term:

- Simplify $A \cap (A \cup B^c)$:

$A \cap (A \cup B^c) = A \quad (\text{absorption law})$

- Simplify $B \cap (A \cup B^c)$:

$B \cap (A \cup B^c) = (B \cap A) \cup (B \cap B^c) \quad (\text{distributy law})$

Since $B \cap B^c = \emptyset$, (Complement law) we have:

$$B \cap (A \cup B^c) = B \cap A$$

3. Combine the simplified terms:

$$(A \cup B) \cap (A \cup B^c) = A \cup (B \cap A)$$

4. Use the absorption law:

$$A \cup (B \cap A) = A$$

Hence,

$$(A \cup B) \cap (A \cup B^c) = A$$

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